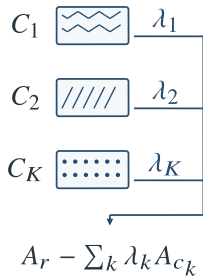


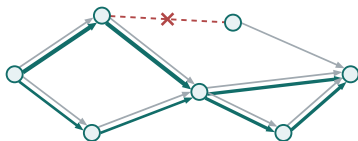
Penalty view

Costs in the objective



SafeOT

$$\langle C_k, F \rangle \leq b_k \quad \forall k$$



Entropic projection $\rightarrow \lambda_k^*$
can price before violation

observed-cost feedback
 $\bar{J}_k - B_k \rightarrow$

One flow, all budgets

b_1

b_2

b_K

state clusters

$\rightarrow$ observed

$\rightarrow$ target

$\cdots$ excluded

clip + EMA

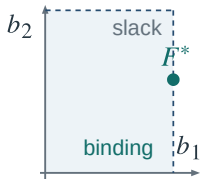
$$\lambda_k = \bar{\lambda}_k^* + \beta_k$$

priced advantage $\rightarrow$ **PPO/TRPO** $\rightarrow \pi_{t+1}$

Constraint view

Budgets bound flow costs

$$\langle C_2, F \rangle$$



$$\langle C_1, F \rangle$$

$$\lambda_1^* > 0, \quad \lambda_2^* = 0$$

One price rule

exact settings

Fixed penalty

$c = 0, \beta$ fixed

Lagrangian

$c = 0, \beta$ adaptive

SafeOT

finite ε

conditional limits

LP

$\varepsilon \rightarrow 0$

Active-set

$\varepsilon \rightarrow \infty$

saturated clip